Opinion Mining from Customer Reviews

Moving beyond simple binary sentiment analysis by extracting **multi-dimensional features** from text using a fine-tuned Large Language Model (LLM).

Sentiment

Polarized classification (Positive/Negative)

Emotions

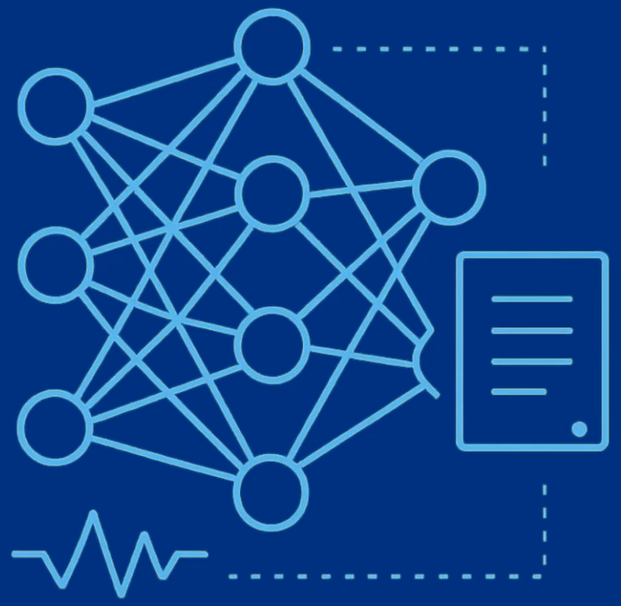
Granular emotional states (e.g., Joy, Disappointment, Excitement)

Aspects

Identifying specific topics mentioned (e.g., Acting, Plot, Visuals, Music)

Intensity

Measuring the strength of the opinion (Strong, Moderate, Weak)



IMDB Dataset & Preprocessing

Data Source

IMDB Large Movie Review Dataset — Raw text reviews paired with structured labels.

Dataset Size

Preprocessing Pipeline

- Applied custom heuristics to derive **emotion** and **intensity** labels from raw ratings and text
- Structured data into a **prompt-response format** suitable for instruction fine-tuning

8,000	800
Training Samples	Testing Samples

Made with GAMMA

Base Model – Qwen2.5-0.5B

Architecture

A **Decoder-only Transformer** model optimized for instruction following and efficiency.

Capabilities

Small footprint (**~500 million parameters**), making it ideal for local fine-tuning and rapid iteration. Pre-trained on diverse multilingual data with a strong grasp of reasoning and text generation.

Baseline Limitations

The base model often struggles with maintaining **strict output formats** and accurately identifying subtle nuances like **Intensity** or specific **Aspects** simultaneously.

Made with GAMMA

Efficiency via Low-Rank Adaptation (LoRA)

Overview

LoRA **freezes the original pre-trained weights** and injects trainable rank-decomposition matrices into the Transformer layers — enabling high-performance adaptation without requiring massive compute resources.

Implementation Details

STEP 1: FREEZE

Freeze pre-trained weights.

STEP 2: INJECT

Inject trainable rank-decomposition matrices into Transformer layers (Attention and MLP).

STEP 3: ADAPT

Achieve high-performance adaptation with dramatically fewer trainable parameters.

Rank (r)

32

Target Layers

Applied to **Attention layers** and **MLP blocks** to capture complex feature relationships

Alpha

64 (scaling factor)

Made with GAMMA

Comparative Metrics & Evaluation Logic

The model is evaluated on **five distinct dimensions** to capture the full complexity of the task. Identical test prompts are run through both the **Base Model** and the **LoRA Model** to quantify improvement.

1

Format Compliance

Does the output follow the exact required syntax?

2

Sentiment Accuracy

Direct comparison of positive vs. negative classification.

3

Emotion Hit Rate

Intersection-over-Union or direct match for emotional labels.

4

Aspect Hit Rate

Accuracy in identifying specific movie features (e.g., acting, pacing).

5

Intensity Accuracy

Correctness of the "Strong / Moderate / Weak" classification.

Made with GAMMA

Performance Analysis: Base vs. LoRA

Base Model

Often fails to adhere to the **multi-field format**, leading to lower Format Compliance scores across the board.

LoRA Model

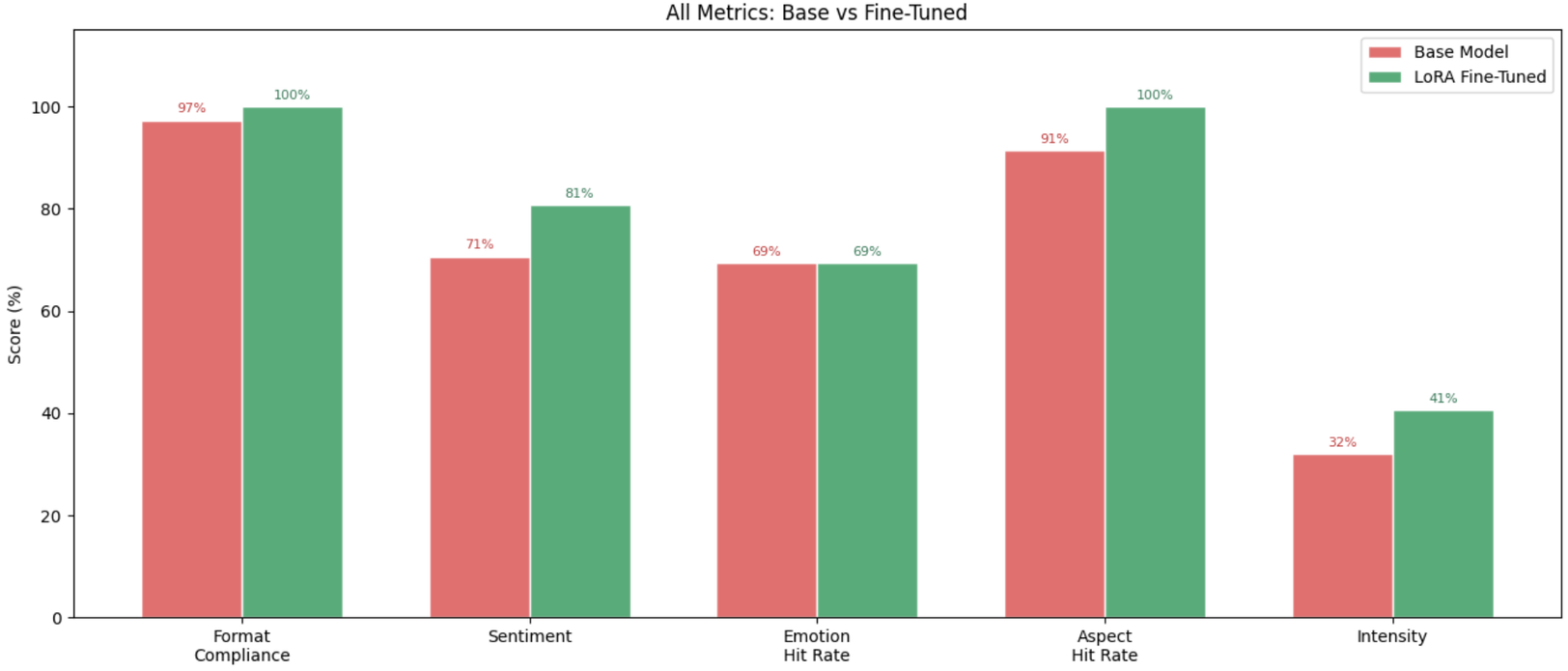
Shows a significant **jump in accuracy** for Aspect and Emotion extraction — the most nuanced dimensions of the task.

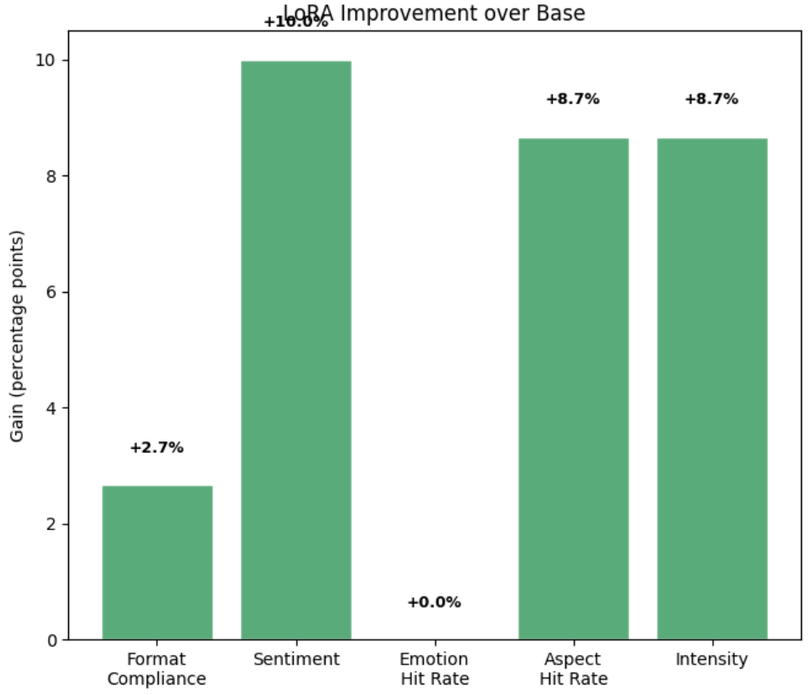
Key Observation

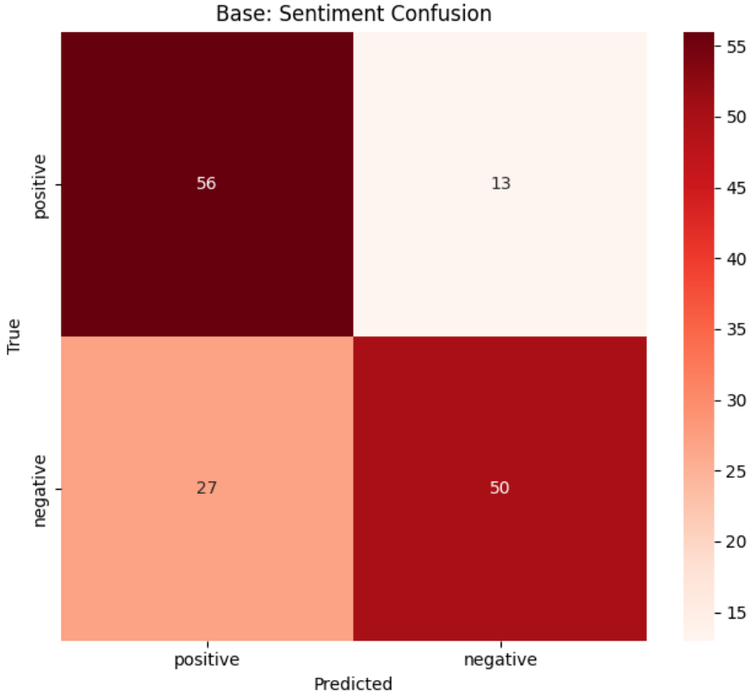
Fine-tuning significantly **reduces hallucinations** and ensures the model provides a structured response even for short or ambiguous reviews.

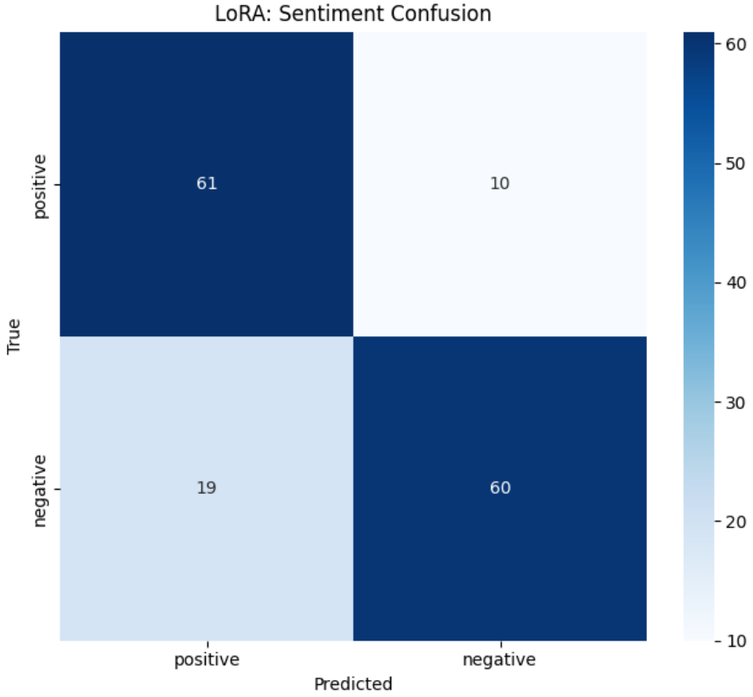
Made with GAMMA

Base Model vs LoRA Fine-Tuned — Feature Extraction Comparison










```
SUMMARY

Format compliance : 97%  →  100%
Sentiment acc.    : 71%  →  81%
Emotion hit rate  : 69%  →  69%
Aspect hit rate   : 91%  →  100%
Intensity acc.    : 32%  →  41%

Train samples : 8,000
Test samples  : 800
LoRA rank     : r=32
Epochs        : 3
```

Showcase: Task-Specific Output Format

Instruction Format

The model is prompted to provide a **single-line structured response** following a strict template.

The Template

```
sentiment: <s> | emotions: <e1>, <e2> | aspects:
<a1>, <a2> | intensity: <i>
```

Example

Input: "The plot twists were shocking and the acting was breathtaking!"

Model Output:

```
sentiment: positive | emotions: surprise, joy |
aspects: plot, acting | intensity: strong
```

Reliability

Demonstrates the ability of the fine-tuned LLM to function as a **reliable data extractor** for downstream analysis — producing consistent, parseable, multi-dimensional opinion signals from unstructured text.

- → Structured, machine-readable output
- → Multi-dimensional feature extraction in one pass
- → Robust even on short or ambiguous reviews

Made with GAMMA

Team Members

Name	ID
Ahmed Ashraf Mohammed	20220009
Aboubakr Ahmed Ali	20220003
Ahmed Alaa Elhousseny	20220026
Ahmed Alaa Fathy	20220027
Ahmed Mahmoud Qassem	20220041

Made with GAMMA